

Q. 1: Tick the correct Answer. (4)

1. The number 2.353535 is a/an:	(A) Irrational number	(B) Rational number	(C) Integer	(D) Natural number
2. The recurring decimal $0.\overline{6}$ is a/an:	(A) Irrational number	(B) Rational number	(C) Prime number	(D) Whole number
3. The sum of three consecutive integers is 42. What are the integers?	(A) 11, 12, 13	(B) 12, 13, 14	(C) 13, 14, 15	(D) 14, 15, 16
4. The markup on Rs. 3,75,000 at a rate of 14% for one year is:	(A) Rs. 37,500	(B) Rs. 14,000	(C) Rs. 52,500	(D) Rs. 42,000

Section-II (Marks:2x8=16)**Q. 2: Write the short answer of any 6 parts of the following. (Marks:2x8=16)**

(i) Represent the following number on number line: $-2\frac{1}{7}$

(ii) Represent the following number on number line: $4\frac{1}{3}$

(iii) Rationalize the denominator of: $\frac{13}{4+\sqrt{3}}$

(iv) Rationalize the denominator of: $\frac{\sqrt{2}-1}{\sqrt{5}}$

(v) Rationalize the denominator of: $\frac{4\sqrt{3}}{\sqrt{7}+\sqrt{5}}$

(vi) Mirha sells a toy for Rs. 1520. What will the selling price be to get a 15% profit?

(vii) The weather in Lahore was unusually warm during the summer of 2024. The TV news reported temperature as high as 48°C . By using the formula, $(^{\circ}\text{F} = \frac{9}{5}^{\circ}\text{C} + 32)$ find the temperature as Fahrenheit scale.

(viii) Express the following recurring decimal as a rational number $\frac{p}{q}$, where p and q are integers: $0.\overline{93}$

Section-III Attempt any 2 Question**Q. 3: Long Question 5x2=10**

(a) Simplify: $\sqrt[7]{\frac{x^{14} \times y^{21} \times z^{35}}{y^{14} z^7}}$

(b) Saleem, Nadeem, and Tanveer earned a profit of Rs. 4,50,000 from a business. If their investments in the business are in the ratio 4: 7: 14, find each person's profit.